MODULE 2

IMAGE TRANSFORMS

- Transform is basically a representation of the signal.
- Image transforms are extensively used in image processing and image analysis.
- Transform is basically a mathematical tool, which allows us to move from one domain to another domain (time domain to the frequency domain).

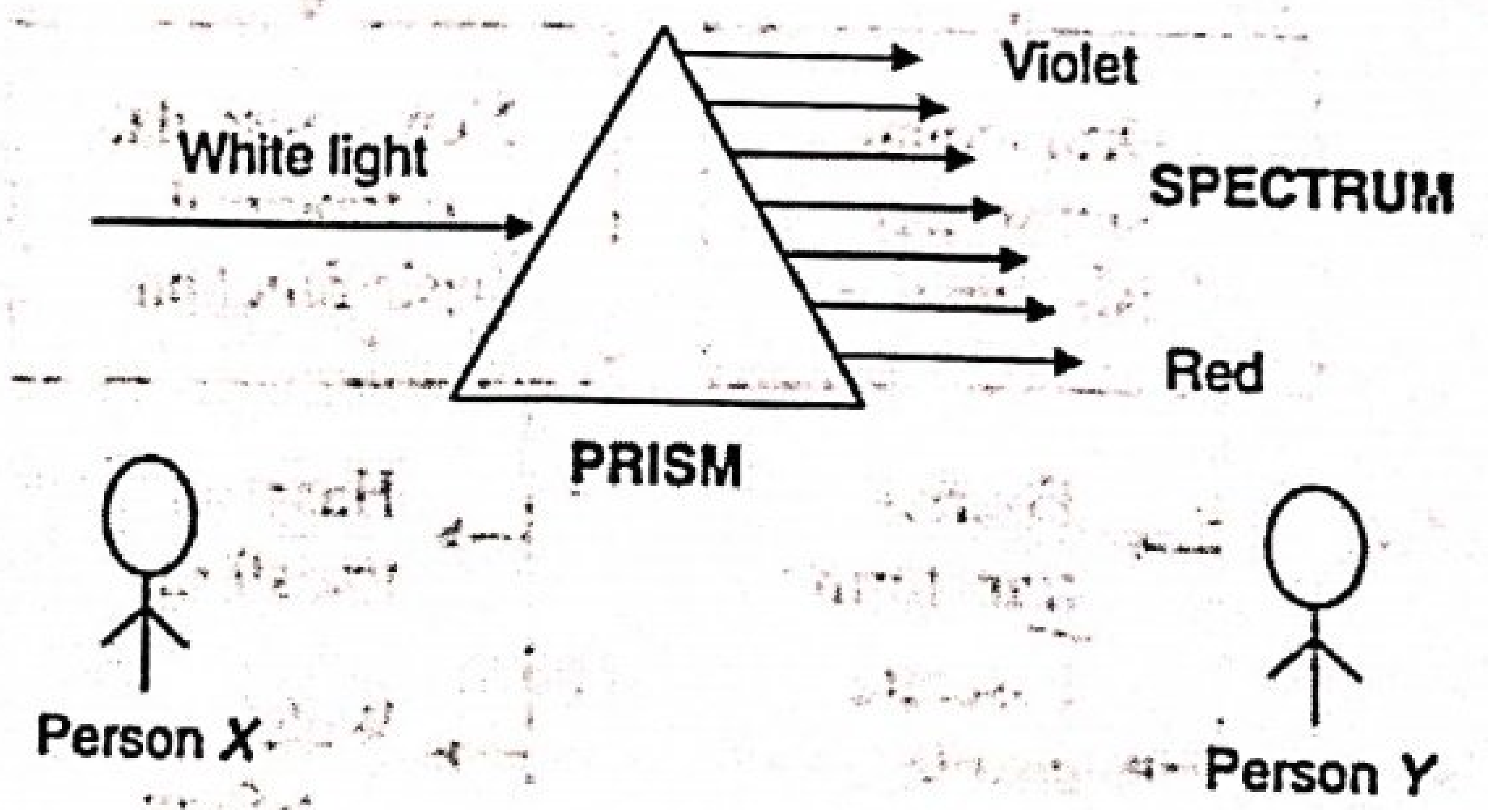
The need for transforms is given as follows:

i) Mathematical Convenience

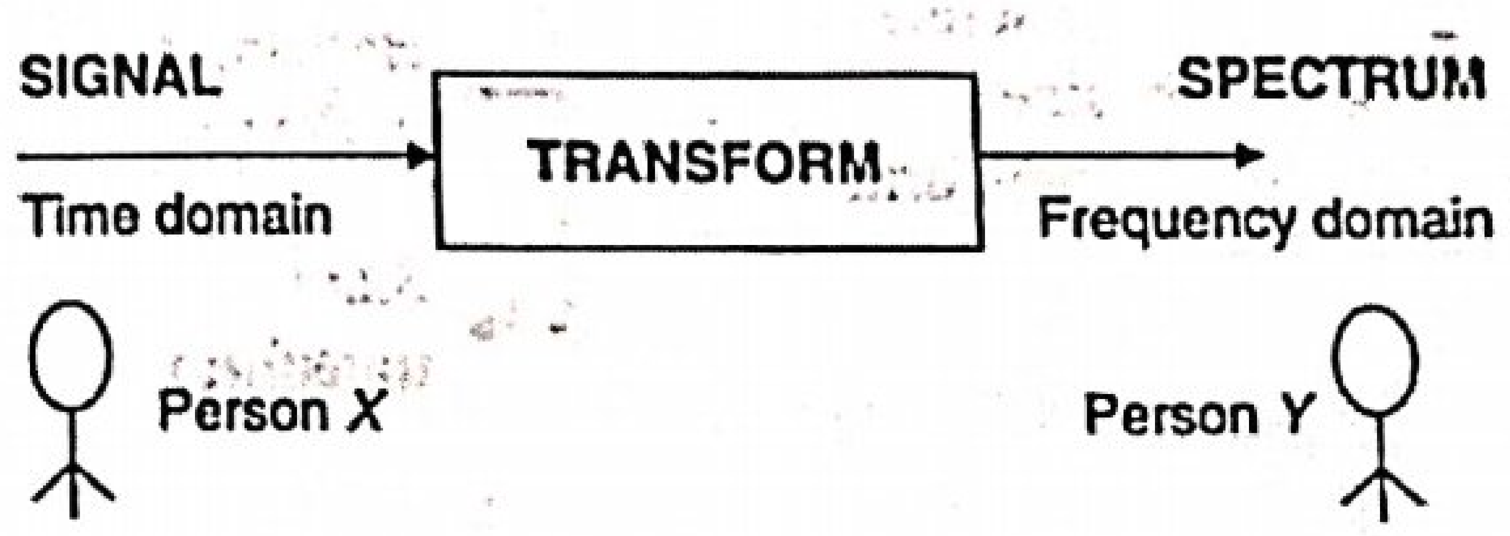
Every action in time domain will have an impact in the frequency domain.

ii) To extract more information

Transforms allows us to extract more relevant information. To illustrate this consider the following example of spectrum of light and concept of transformation.



Spectrum of white light



Concept of transformation

Basic Concept Of Spatial Domain and Frequency Domain

Spatial Domain:

- Enhancement of the image space that divides an image into uniform pixels according to the spatial coordinates with a particular resolution.
- The spatial domain methods perform operation on pixels directly.

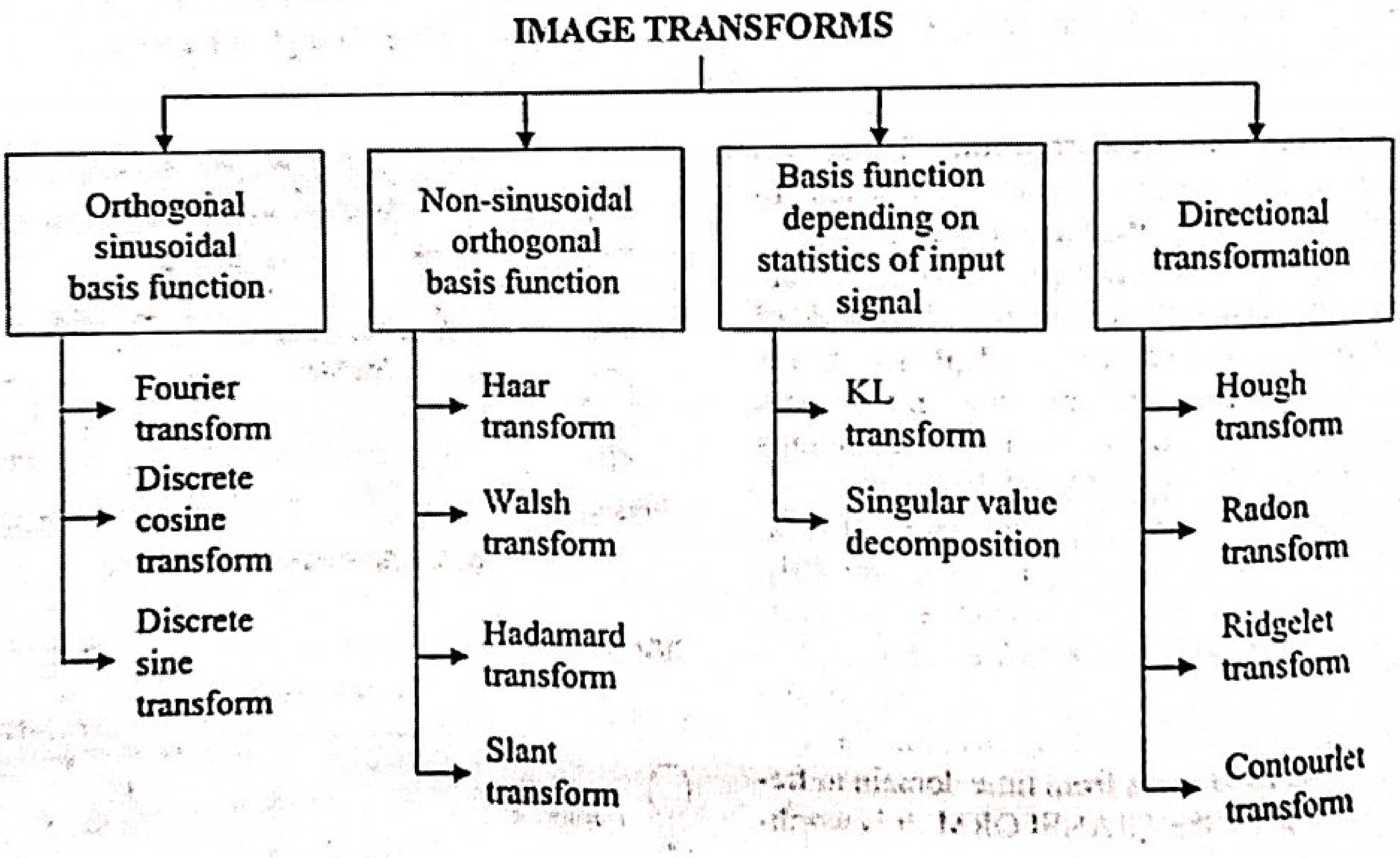
Frequency Domain:

- The frequency domain refers to the analytic space in which mathematical functions or signals are conveyed in terms of frequency rather than time.
- Frequency domain enhancement may be obtained by applying the fourier transform to the spatial domain.

* Spatial domain (image plane)
  - Techniques are based on directly manipulation of pixels in an image.

* Frequency domain
  - Techniques are based on modifying the fourier transform of an image.

Classification of Image Transforms.



UNITARY TRANSFORMS

A unitary transformation preserves the signal energy. For most image processing applications anyone of the mathematical transformation are applied to the signal or images to obtain further information from that signal.

1. One Dimensional Signals.

For an one dimensional sequence $\{f(x), 0 \leq x \leq N-1\}$ represented as a vector

$f = [f(0) f(1) \quad f(N-1)]^T$ of size N, a transformation may be written as

$$g = T \cdot f \Rightarrow g(u) = \sum_{x=0}^{N-1} T(u,x) f(x), 0 \leq u \leq N-1$$

Where $g(u)$ is the transform (or transformation) of $f(x)$ and

$T(u,x)$ is the so called forward transformation Kernel.

Similarly, the inverse transform is the relation

$$f(x) = \sum_{u=0}^{N-1} I(x,u) g(u), 0 \leq x \leq N-1$$

Or written in a matrix form

$$f = I \cdot g = T^{-1} \cdot g.$$

Where $I(x,u)$ is the so called inverse transformation kernel.

If $\quad I = T^{-1} = T^{*T}$

the matrix T is called unitary, and the transformation is called unitary as well.

It can be proven that the columns (or rows) of an $N \times N$ unitary matrix are orthogonal and therefore, form a complete set of basis vectors in the N-dimensional vector space.

$$f = T^{*T} \cdot g \Rightarrow f(x) = \sum_{u=0}^{N-1} T^{*}(u,x)\, g(u)$$

The column T^{*T}, that is the vectors

$T^{*}u = [T^{*}(u,0)\ T^{*}(u,1)\ T^{*}(u,N-1)]$ are called the basis vectors of T.

Two Dimensional Signals (Images)

As a one dimensional signal can be represented by an orthogonal set of basis vectors, an image can also be expanded in terms of a discrete set of basis arrays called basis images through a two dimensional (image) transform.

For an NxN image $f(x,y)$ the forward and inverse transforms are given below.

$$g(u,v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T(u,v,x,y)\, f(x,y)$$

$$f(x,y) = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} I(x,y,u,v)\, g(u,v)$$

Where $T(u,v,x,y)$ and $I(x,y,u,v)$ are called the forward and inverse transformation kernels respectively.

It is said to be :

<u>Separable</u> if $T(u,v,x,y) = T_1(u,x)\, T_2(v,y)$

<u>Symmetric</u> if T_1 is functionally equal to T_2 such that $T(u,v,x,y) = T_1(u,x)\, T_1(v,y)$

The same comments are valid for the <u>inverse</u> <u>kernel</u>

If the kernel $T(u,v,x,y)$ of an image transform is separable and symmetric, then the transform

$$g(u,v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T(u,v,x,y) f(x,y)$$

$$= \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T_1(u,x) T_1(v,y) f(x,y)$$ can be written

in matrix form as follows.

$$g = T_1 \cdot f \cdot T_1^T$$

Where f is the original image of size $N \times N$.

T_1 is the transformation matrix with

elements $t_{ij} = T_1(i,j)$

If, T_1 is a unitary matrix then the transform is called separable unitary and the original image is recovered through the relationship.

$$f = T_1^{*T} \cdot g \cdot T_1^*$$

Fundamental Properties of unitary transforms.

1. The property of energy preservation.

In the unitary transformation

$$g = T \cdot f$$

It is easily proven that $|g|^2 = |f|^2$ $[T^{-1} = T^{*T}]$

Thus, a unitary transformation preserves the signal energy. This property is called energy preservation property.

This means that every unitary transformation is simply a rotation of the vector f in the N-dimensional vector space.

For the 2D case the energy preservation can be written as:

$$\sum_{x=0}^{N-1} \sum_{y=0}^{N-1} |f(x,y)|^2 = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} |g(u,v)|^2$$

2. The property of energy compaction

Most unitary transforms pack a large fraction of the energy of the image into relatively few of the transform coefficients. This means that relatively few of the transform coefficients have significant values and these are the coefficient that are close to the origin. This property is very useful for compression purposes.

DISCRETE FOURIER TRANSFORM (DFT)

The Fourier transform was developed by Jean Baptiste Joseph Fourier to explain the distribution of temperature and heat conduction.

One way to analyze spatial variation is to decompose an image into a set of orthogonal functions, one such set being the Fourier transforms. A Fourier transform is used to transform an intensity image into the domain of spatial frequency.

For a continuous time signal $x(t)$, the Fourier transform is defined as $X(\Omega)$

$$x(t) \xrightarrow{CTFT} X(\Omega)$$

Continuous Time Fourier Transform (CTFT) is defined as

$$X(\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t}\, dt \quad \text{———} (1)$$

A continuous time signal $x(t)$ is converted into discrete time signal $x(nT)$ by sampling process, where T is the sampling interval.

$$x(t) \xrightarrow{\text{sampling}} x(nT) \quad \text{———} (2)$$

The Fourier transform of a finite energy discrete time Signal $x(nT)$ is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(nT)\, e^{-j\Omega nT} \quad \text{———— (3)}$$

Where $X(e^{j\omega})$ is known as Discrete-Time Fourier Transf (DTFT) and is continuous function of ω.

The relation between ω & Ω is given by.

$$\omega = \Omega T \quad \text{———— (4)}$$

Replacing Ω by $2\pi f$

$$\omega = 2\pi f T. \quad \text{———— (5)}$$

Wher T is the sampling interval and is equal to $\frac{1}{f_s}$. Replacing T by $\frac{1}{f_s}$

$$\omega = 2\pi f \times \frac{1}{f_s} \quad \text{———— (6)}$$

$$\frac{f}{f_s} = k \quad \text{———— (7)}$$

By Substituting (7) in (6).

$$\omega = k \times 2\pi \quad \text{———— (8)}$$

Eqn (8) is modified as.

$$\frac{\omega}{2\pi} = \frac{k}{N} \quad \text{———} (9)$$

The DFT of a finite duration sequence $x(n)$ is defined as

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}kn} \quad \text{———} (10)$$

where $k = 0, 1, \ldots N-1$.

UNITARY TRANSFORM

A discrete linear transform is unitary if its transform matrix conforms to the unitary condition

$$A \times A^H = I$$

Where A = transformation matrix, A^H represents Hermitian matrix.

$$A^H = A^{*T}$$

I = Identity matrix

When the transform matrix A is unitary, the defined transform is called unitary transform.

Q) Check whether the DFT matrix is unitary or not.

Soln:

Step 1: Determination of the matrix A

Finding 4-point DFT (where N=4)

The formula to compute a DFT matrix of order 4 is given below.

$$X(k) = \sum_{n=0}^{3} x(n)\, e^{-j\frac{2\pi}{4}kn} \quad \text{where } k=0,1,2,3.$$

1. Finding $X(0)$

$$X(0) = \sum_{n=0}^{3} x(0) + x(1) + x(2) + x(3).$$

2. Finding $X(1)$

$$X(1) = \sum_{n=0}^{3} x(n)\, e^{-j\frac{2\pi}{4}kn}$$

$$= x(0) + x(1)\, e^{-j\frac{\pi}{2}} + x(2)\, e^{-j\pi} + x(3)\, e^{-j\frac{3\pi}{2}}$$

$$X(1) = x(0) - j\,x(1) - x(2) + j\,x(3).$$

3. Finding $X(2)$

$$X(2) = \sum_{n=0}^{3} x(n)\, e^{-j\pi} + x(2)\, e^{-j2\pi} + x(3)\, e^{-j3\pi}$$

$$X(2) = x(0) - x(1) + x(2) - x(3).$$

4. Finding $X(3)$

$$X(3) = \sum_{n=0}^{3} x(n)\, e^{-j\frac{3\pi}{2}n}$$

$$= x(0) + x(1)e^{-j\frac{3\pi}{2}} + x(2)e^{-j3\pi} + x(3)e^{-j\frac{9\pi}{2}}$$

$$X(3) = x(0) + j\,x(1) - x(2) - j\,x(3).$$

Collecting the coefficients of $X(0)$, $X(1)$, $X(2)$ and $X(3)$, we get

$$X[k] = A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

Step 2: Computation of A^H

To determine A^H, first determine the conjugate and then take its transpose.

$$A \xrightarrow{\text{Conjugate}} A^* \xrightarrow{\text{Transpose}} A^H$$

Step 2a Computation of conjugate A^*

$$A^* = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

Step 2b Determination of transpose of A^*

$$(A^*)^T = A^H = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{vmatrix}$$

Step 3 Determination of $A \times A^H$

$$A \times A^H = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{vmatrix}$$

$$= \begin{vmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{vmatrix} = 4 \times \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

The result is the identity matrix, which shows that Fourier transform satisfies unitary condition.

Sequency

Sequency refers to the number of sign change. The sequency for a DFT matrix of order 4 is given below.

Transform coefficients:

$$x[k] = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{vmatrix} \begin{array}{l} \textbf{SEQUENCY} \\ \rightarrow \text{Zero sign change} \\ \rightarrow \text{Two sign changes} \\ \rightarrow \text{Three sign change} \\ \rightarrow \text{One sign change} \end{array}$$

2D DISCRETE FOURIER TRANSFORM (DFT)

The 2D-DFT of a rectangular image $f(m,n)$ of size $M \times N$ is represented as $F(k,l)$

$$f(m,n) \xrightarrow{\text{2D-DFT}} F(k,l)$$

Where $F(k,l)$ is defined as

$$F(k,l) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m,n)\, e^{-j\frac{2\pi}{M}mk}\, e^{-j\frac{2\pi}{N}nk}$$

$$F(k,l) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m,n)\, e^{-j\frac{2\pi}{M}nl}\, e^{-j\frac{2\pi}{N}mk}$$

For a square image $f(m,n)$ of size $N \times N$, the 2D DFT is defined as.

$$F(k,l) = \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} f(m,n) e^{-j\frac{2\pi}{N}mk} e^{-j\frac{2\pi}{N}nl}$$

The inverse 2D Discrete Fourier Transform is given by

$$f(m,n) = \frac{1}{N^2} \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} F(k,l) e^{j\frac{2\pi}{N}mk} e^{j\frac{2\pi}{N}nl}$$

The Fourier transform $F(k,l)$ is given by

$$F(k,l) = R(k,l) + jI(k,l)$$

Where $R(k,l)$ represents the real part of the spectrum and $I(k,l)$ represents the imaginary part.

The Fourier transform $F(k,l)$ can be expressed in polar coordinate as

$$F(k,l) = |F(k,l)| e^{j\varphi(k,l)}$$

Where $|F(k,l)| = \left(R^2\{F(k,l)\} + I^2\{F(k,l)\}\right)^{1/2}$ is called the magnitude spectrum of the Fourier transform and $\varphi(k,l) = \tan^{-1}\frac{I\{F(k,l)\}}{R\{F(k,l)\}}$ is the phase angle or spectrum.

Here $R\{F(k,l)\}$, $I\{F(k,l)\}$ are the real and imaginary parts of $F(k,l)$ respectively.

The frequency range is given by

$$[0,N]\times[0,M]$$

When M is the horizontal resolution of the image.

N is the vertical resolution of the image.

In the case of a digital image, the spectrum will be unique in the range $-\pi$ to π or between 0 to 2π.

Properties of 2D DFT

Property	Sequence	Transform
Spatial shift	$f(m-m_0, n)$	$e^{-j\frac{2\pi}{N}m_0 k}F(k,l)$
Periodicity	–	$F(k+pN, l+qN) = F(k,l)$
Convolution	$f(m,n)*g(m,n)$	$F(k,l)\times G(k,l)$
Scaling	$f(am, bn)$	$\frac{1}{\lvert ab\rvert}F(k/a, l/b)$
Conjugate symmetry		$F(k,l) = F^*(-k,-l)$
Multiplication by exponential	$e^{j\frac{2\pi}{N}mk_0}e^{j\frac{2\pi}{N}nl_0}f(m,n)$	$F(k-k_0, l-l_0)$
Rotation property	$f(r\cos(\theta+\theta_0), r\sin(\theta+\theta_0))$	$F[R\cos(\Phi+\Phi_0), R\sin(\Phi+\Phi_0)]$
Folding property	$f(-m,-n)$	$F(-k,-l)$

Q> Compute the 2D DFT of the 4×4 gray scale image given below.

$$f[m,n] = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Solution:-

The 2D DFT of the image $f[m,n]$ is represented as $F[k,l]$.

$$F[k,l] = \text{Kernel} \times f[m,n] \times (\text{kernel})^T \quad \text{——— (1)}$$

The Kernel or basis of the Fourier transform for $N=4$ is given by

The DFT basis for $N=4$ is given by $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$ ——(2)

Substituting Eq.(2) in (1), we get

$$F(k,l) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

upon simplification, we get

$$F(k,l) = \begin{bmatrix} 4 & 4 & 4 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$= \begin{bmatrix} 16 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Q> Compute the inverse 2D DFT of the transform coefficients given by.

$$F(k,l) = \begin{bmatrix} 16 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Solution:-

The inverse 2D DFT of the fourier coefficients $F[k,l]$ is given by $f[m,n]$ as

$$f[m,n] = \frac{1}{N^2} \times \text{kernel} \times F[k,l] \times (\text{kernel})^T$$

In this example, $N=4$.

$$f[m,n] = \frac{1}{16} \times \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \times \begin{bmatrix} 16 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$f[m,n] = \frac{1}{16} \times \begin{bmatrix} 16 & 0 & 0 & 0 \\ 16 & 0 & 0 & 0 \\ 16 & 0 & 0 & 0 \\ 16 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$f[m,n] = \frac{1}{16} \times \begin{bmatrix} 16 & 16 & 16 & 16 \\ 16 & 16 & 16 & 16 \\ 16 & 16 & 16 & 16 \\ 16 & 16 & 16 & 16 \end{bmatrix}$$

$$f[m,n] = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

The Fourier transform $F(k,l)$ can be expressed in polar coordinate as $F(k,l) = |F(k,l)| e^{j\phi(k,l)}$

Where $|F(k,l)| = (R^2\{F(k,l)\} + I^2\{F(k,l)\})^{1/2}$ is called the magnitude spectrum of the Fourier transform and

$\phi(k,l) = \tan^{-1} \dfrac{I\{F(k,l)\}}{R\{F(k,l)\}}$ is the phase angle or phase spectrum

Here, $R\{F(k,l)\}$, $I\{F(k,l)\}$ are the real and imaginary parts of $F(k,l)$ respectively.

DISCRETE COSINE TRANSFORM [DCT]

★ The Discrete Cosine Transform was developed by Ahmed, Natrajan and Rao in 1974. A discrete cosine transform consists of a set of basis vectors that are sampled cosine functions.

★ DCT is a technique for converting a signal into elementary frequency components and it is widely used in image compression.

If $x[n]$ is the signal of length N, the Fourier transform of the signal $x[n]$ is given by $X[k]$.

Where $$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi kn}{N}}$$

where k varies between 0 to N-1.

Now consider the extension of the signal $x[n]$ which is denoted by $x_e[n]$ so that the length of the extended sequence is 2N. The sequence $x[n]$ can be extended in two ways.

Let us consider a sequence (original sequence) of length four given by $x[n] = [1, 2, 3, 4]$. The origin sequence and extended sequences are shown in the following Figures Fig 1 and Fig 2.

The extended sequence can be produced by simply copying the original sequence as shown in Fig 2.

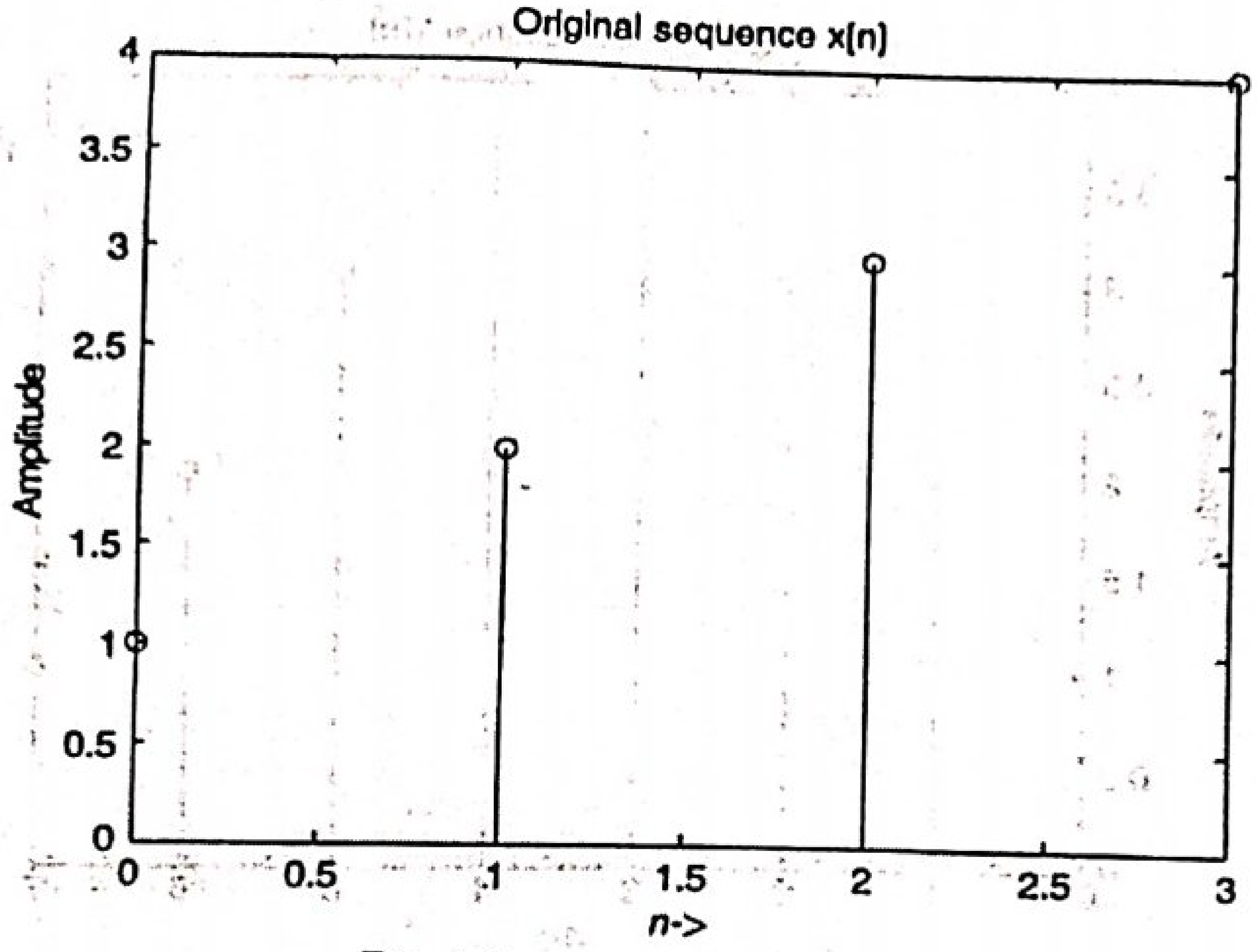


Fig 1 : Original sequence

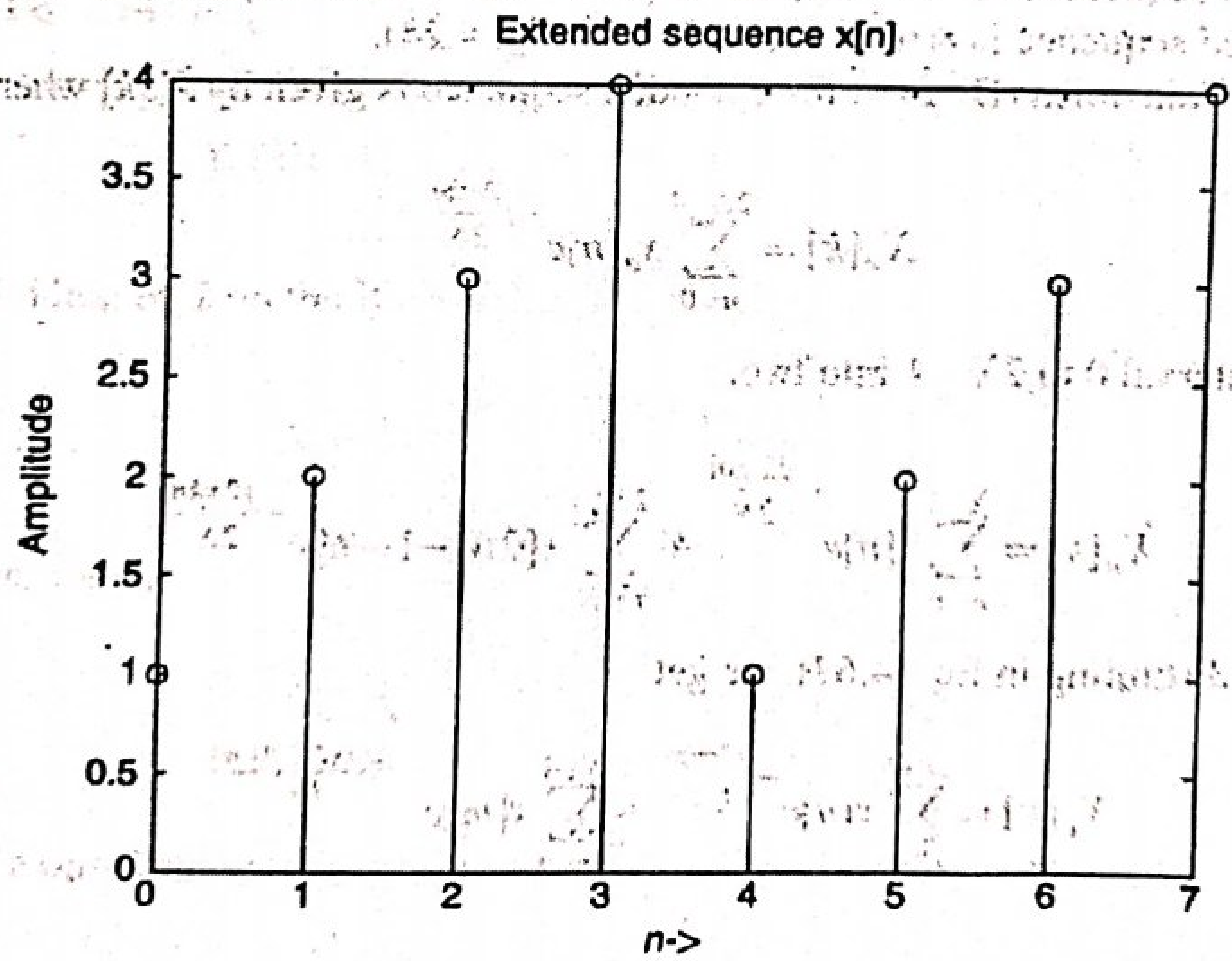


Fig 2 : Extended sequence obtained by simply copying the original sequence

The main drawback of the extended method is the variation in the value of the sample at $n=3$ and at $n=4$. To overcome this, a second method of obtaining the extended sequence is by copying the original sequence in a folded manner as shown in Figure 3.

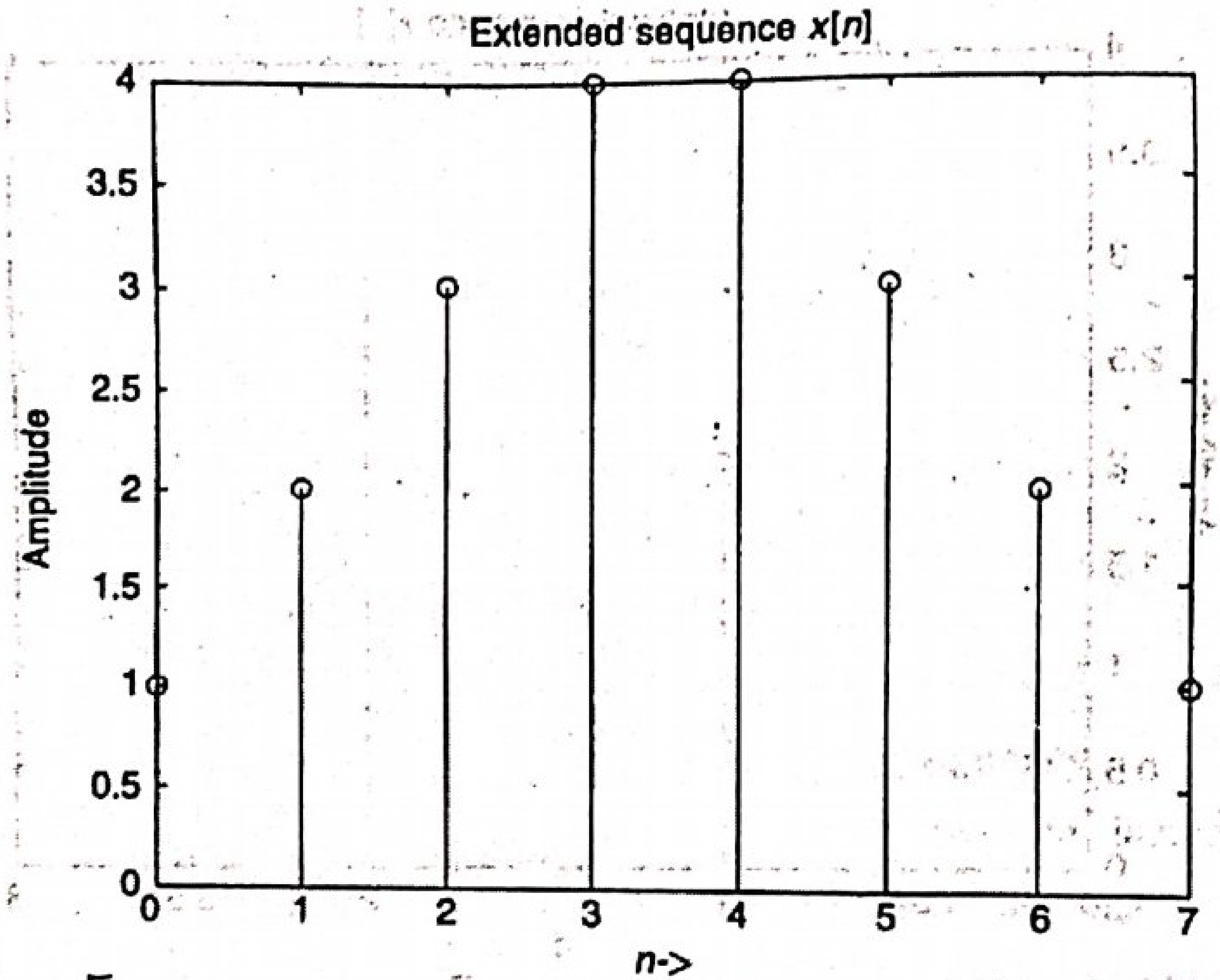


Fig. 3 : *Extended sequence obtained by folding the original sequence*

The Discrete Fourier Transform (DFT) of the extended sequence is given by $X_e[k]$ where

$$X_e[k] = \sum_{n=0}^{2N-1} x_e[n]\, e^{-j\frac{2\pi kn}{2N}}$$

Now, we can split the interval 0 to $2N-1$ into two.

$$X_e[k] = \sum_{n=0}^{N-1} x[n]\, e^{\frac{-j2\pi kn}{2N}} + \sum_{n=N}^{2N-1} x[2N-1-n]\, e^{\frac{-j2\pi kn}{2N}}$$

{As per the syllabus of CST438 Image Processing Techniques derivation of DCT is not required}

Upon simplification, we get

$$\frac{X_e[k]\, e^{\frac{-j\pi k}{2N}}}{2} = \sum_{n=0}^{N-1} x[n] \cos\left\{\frac{(2n+1)\pi k}{2N}\right\}$$

Thus the kernel of a one-dimensional discrete cosine transform is given by

$$X[k] = \alpha(k) \sum_{n=0}^{N-1} x[n] \cos\left\{\frac{(2n+1)\pi k}{2N}\right\},$$

where $0 \leq k \leq N-1$.

$$\alpha(k) = \sqrt{\frac{1}{N}} \text{ if } k=0.$$

$$\alpha(k) = \sqrt{\frac{2}{N}} \text{ if } k \neq 0$$

One dimensional discrete cosine transform is given by

$$x[n] = \alpha(k) \sum_{k=0}^{N-1} X[k] \cos\left[\frac{(2n+1)\pi k}{2N}\right], \quad 0 \leq n \leq N-1.$$

The forward 2D DCT of a signal $f(m,n)$ is given by

$$F[k,l] = \alpha(k)\alpha(l)\sum_{m=0}^{N-1}\sum_{n=0}^{N-1} f(m,n)\cos\left[\frac{(2m+1)\pi k}{2N}\right]\cos\left[\frac{(2n+1)\pi l}{2N}\right]$$

Where
$$\alpha(k) = \begin{cases}\sqrt{\frac{1}{N}} & \text{if } k=0\\ \sqrt{\frac{2}{N}} & \text{if } k\neq 0\end{cases}$$

Similarly
$$\alpha(l) = \begin{cases}\sqrt{\frac{1}{N}} & \text{if } l=0\\ \sqrt{\frac{2}{N}} & \text{if } l\neq 0\end{cases}$$

The 2D inverse DCT (IDCT) is given by.

$$f(m,n) = \sum_{k=0}^{N-1}\sum_{l=0}^{N-1}\alpha(k)\alpha(l)F(k,l)\cos\left[\frac{(2m+1)\pi k}{2N}\right]\cos\left[\frac{(2n+1)\pi k}{2N}\right]$$

2D-IDCT

$$F(m,n) = \sum_{k=0}^{N-1}\sum_{l=0}^{N-1}\alpha(k)\alpha(l)F(k,l)\cos\left[\frac{(2m+1)\pi k}{2N}\right]\cos\left[\frac{(2n+1)\pi l}{2N}\right]$$

2D-DCT

$$F(k,l) = \alpha(k)\alpha(l)\sum_{m=0}^{N-1}\sum_{n=0}^{N-1} f(m,n)\cos\left[\frac{(2m+1)\pi k}{2N}\right]\cos\left[\frac{(2n+1)\pi l}{2N}\right]$$

Q) Compute the discrete cosine transform (DCT) matrix for N=4.

Solution:

$$X[k] = \alpha(k) \sum_{n=0}^{N-1} x[n] \cos\left\{\frac{(2n+1)\pi k}{2N}\right\}, \text{where } 0 \le k \le N-1$$

Where

$$\alpha(k) = \sqrt{\frac{1}{N}} \quad \text{if } k=0$$

$$\alpha(k) = \sqrt{\frac{2}{N}} \quad \text{if } k \neq 0$$

In this case, the value of N=4. Substituting N=4 in the expression of X[k] we get

$$X[k] = \alpha(k) \sum_{n=0}^{3} x[n] \cos\left[\frac{(2n+1)\pi k}{8}\right] \quad \text{—(1)}$$

Substituting k=0 in Eq(1), we get

$$X[0] = \sqrt{\frac{1}{4}} \times \sum_{n=0}^{3} x[n] \cos\left[\frac{(2n+1)\pi \times 0}{8}\right]$$

$$= \frac{1}{2} \times \sum_{n=0}^{3} x[n] \cos(0)$$

$$= \frac{1}{2} \times \sum_{n=0}^{3} x[n] \times 1$$

$$= \frac{1}{2} \times \sum_{n=0}^{3} x[n]$$

$$= \frac{1}{2} \times \left\{ x(0) + x(1) + x(2) + x(3) \right\}$$

$$X[0] = \frac{1}{2} x(0) + \frac{1}{2} x(1) + \frac{1}{2} x(2) + \frac{1}{2} x(3).$$

Substituting $k = 1$ in Eq(1), we get.

$$X[1] = \sqrt{\frac{2}{4}} \times \sum_{n=0}^{3} x[n] \cos\left[\frac{(2n+1)\pi \times 1}{8}\right]$$

$$= \sqrt{\frac{1}{2}} \times \sum_{n=0}^{3} x[n] \cos\left[\frac{(2n+1)\pi}{8}\right]$$

$$= \sqrt{\frac{1}{2}} \times \left\{ x(0) \cos\left[\frac{\pi}{8}\right] + x(1) \cos\left(\frac{3\pi}{8}\right) + x(2) \cos\left(\frac{5\pi}{8}\right) + x(3) \cos\left(\frac{7\pi}{8}\right) \right\}$$

$$= 0.707 \times \{ x(0) \times 0.9239 + x(1) \times 0.3827 + x(2) \times (-0.3827) + x(3) \times (-0.9239) \}.$$

$$X(1) = 0.6532\, x(0) + 0.2706\, x(1) - 0.2706\, x(2) - 0.6532\, x(3)$$

Substituting $k=2$ in Eq (1), we get

$$X(2) = \sqrt{\frac{2}{4}} \sum_{n=0}^{3} x[n] \cos\left[\frac{(2n+1)\pi \times 1}{8}\right]$$

$$= \sqrt{\frac{1}{2}} \times \sum_{n=0}^{3} x[n] \cos\left[\frac{(2n+1)\pi}{8}\right]$$

$$= \sqrt{\frac{1}{\sqrt{2}}} \times \left\{ x(0) \times \cos\left(\frac{\pi}{4}\right) + x(1) \times \cos\left(\frac{3\pi}{4}\right) + x(2) \cos\left(\frac{5\pi}{4}\right) + x(3) \times \cos\left(\frac{7\pi}{4}\right) \right\}$$

$$= 0.707 \times \{ x(0) \times 0.7071 + x(1) \times -0.7071 + x(2)(-0.7071) + x(3)(0.7071) \}$$

$$X(2) = 0.5\, x(0) - 0.5\, x(1) - 0.5\, (x(2) + 0.5\, x(3)$$

Substituting $k=3$ in Eq (1), we get

$$X(3) = \sqrt{\frac{2}{4}} \sum_{n=0}^{3} x[n] \cos\left[\frac{(2n+1)\pi \times 3}{8}\right]$$

$$= \sqrt{\frac{1}{2}} \times \sum_{n=0}^{3} x[n] \cos\left[\frac{(2n+1) \times 3\pi}{8}\right]$$

$$= \sqrt{\frac{1}{2}} \times \left\{ x(0) \times \cos\left(\frac{3\pi}{8}\right) + x(1) \cos\left(\frac{9\pi}{8}\right) + x(2) \cos\left(\frac{15\pi}{8}\right) + x(3) \cos\left(\frac{21\pi}{8}\right) \right\}$$

$$= 0.7071 \times \{ x(0) \times 0.3827 + x(1) \times (-0.9239) + x(2) \times (0.9239) + x(3) \times (-0.3827) \}$$

$$X(3) = 0.2706\, x(0) - 0.6533\, x(1) + 0.6533\, x(2) - 0.2706\, x(3)$$

Collecting the coefficients of $x(0), x(1), x(2)$ and $x(3)$ from $X(0), X(1), X(2)$ and $X(3)$, we get

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.6532 & 0.2706 & -0.2706 & -0.6532 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.2706 & -0.6533 & 0.6533 & -0.2706 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

Properties of Discrete Cosine Transform.

The discrete cosine transform is real and orthogonal

If A is a DCT matrix of order N, and if the matrix A is orthogonal then

$$A \times A^T = I$$

Let A be a DCT matrix of order four (4). The matrix A is given by.

$$A = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.6532 & 0.2706 & -0.2706 & -0.6532 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.2706 & -0.6533 & 0.6533 & -0.2706 \end{bmatrix}$$

Then $$A^T = \begin{bmatrix} 0.5 & 0.6532 & 0.5 & 0.2706 \\ 0.5 & 0.2706 & -0.5 & -0.6533 \\ 0.5 & -0.2706 & -0.5 & 0.6533 \\ 0.5 & -0.6532 & 0.5 & -0.2706 \end{bmatrix}.$$

$$A \times A^T = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.6532 & 0.2706 & -0.2706 & -0.6532 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.2706 & -0.6533 & 0.6533 & -0.2706 \end{bmatrix} \begin{bmatrix} 0.5 & 0.6532 & 0.5 & 0.2706 \\ 0.5 & 0.2706 & -0.5 & -0.6533 \\ 0.5 & -0.2706 & -0.5 & 0.6533 \\ 0.5 & -0.6532 & 0.5 & 0.2706 \end{bmatrix}$$

$$A \times A^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I$$

The above result implies $A^{-1} = A^T$.

HADAMARD TRANSFORM

The Hadamard transform is basically the same as the Walsh transform except the rows of the transform matrix are re-ordered. The elements of the mutually orthogonal basis vectors of a Hadamard transform are either +1 or -1, which results in very low computational complexity in the calculation of the transform coefficients. Hadamard matrices are easily constructed for $N = 2^n$ by the following procedure

The order N=2 Hadamard matrix is given as

$$H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \text{———} \quad (1)$$

The Hadamard matrix of order 2N can be generated by kronecker product operation:

$$H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix} \quad \text{———} \quad (2)$$

$$H_4 = \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \quad \text{———} \quad (3)$$

Similarly, substituting, $N = 4$ in (2), we get

$$H_8 = \begin{bmatrix} H_4 & H_4 \\ H_4 & -H_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{bmatrix}$$

The Hadamard matrix of order $N = 2^n$ may be generated from the order two core matrix. It is not desirable to store the entire matrix.